

Liability as a Requirement: Allocating Civil Responsibility for Agentic AI Systems

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Abstract – The proliferation of agentic artificial intelligence (AI) systems autonomous, goal-directed agents capable of acting with minimal human oversight, has exposed fundamental inadequacies in existing civil liability frameworks. Traditional tort and product liability regimes, premised on predictable causal chains and identifiable human actors, struggle to accommodate systems whose emergent behavior arises from complex, and opaque processes. This paper argues that civil liability allocation should be reconceived as a legal requirements specification problem, with the deployer emerging as the primary locus of accountability. Drawing on comparative civil law analysis and the evolving European regulatory framework, we propose a structured set of deployer-oriented legal requirements—encompassing duties of explainability, and risk coverage that can anchor a coherent liability regime and embed trustworthiness into agentic AI design from the outset.

Keywords – *agentic AI, civil liability, requirements engineering, deployer responsibility, trustworthy AI, product liability, tort law*

I. INTRODUCTION

Agentic artificial intelligence systems are no longer speculative. Autonomous agents that plan, retrieve information, invoke external tools, and execute multi-step tasks with minimal human intervention are now deployed in high-stakes domains including healthcare, legal services, financial advisory, and critical infrastructure. Unlike traditional software, which executes predefined instructions, agentic AI systems exhibit goal-directed behaviour that emerges from the interaction of foundation models, retrieval mechanisms, and environmental feedback. This emergent character generates what Asaro has termed the “liability problem”: no natural human actor in the AI supply chain can be straightforwardly identified as the proximate cause of harm [1].

The question of civil liability for agentic AI harm is not merely a doctrinal puzzle. It is a design imperative. Legal accountability structures shape the incentives of deployers, and users alike. If liability is ambiguously allocated, or effectively unallocable, the result is a governance vacuum that leaves harm victims without redress and removes *ex ante* incentives for safety investment. Conversely, a well-specified liability framework functions as a set of binding legal requirements that drive trustworthy system design.

This paper makes three contributions. First, it maps the structural features of agentic AI that render traditional liability

frameworks inadequate. Second, it advances a deployer-centred liability hypothesis, arguing that the entity that puts an agentic system into operational use is best positioned to bear primary civil responsibility. Third, it reframes civil liability obligations as a structured set of legal requirements that should inform the testing, and deployment of trustworthy agentic AI systems. The paper proceeds as follows: Section II characterises the technical features of agentic AI that generate legal challenges; Section III examines the failure of traditional liability regimes; Section IV develops the deployer liability hypothesis; Section V translates it into specifiable legal requirements; and Sections VI and VII discuss implications and conclude.

II. AGENTIC AI: TECHNICAL CHARACTERISTICS GENERATING LEGAL CHALLENGES

The term “agentic AI” denotes systems that combine large language model capabilities with persistent memory, planning modules, and the capacity to invoke external tools or other sub-agents to accomplish user-defined goals across extended task horizons [2]. Unlike single-shot inference, agentic pipelines compose capabilities dynamically in response to task context, producing sequences of actions whose combined effects may be difficult to foresee at design time [3].

Four technical characteristics create acute legal challenges. First, emergent behaviour: agentic systems generate actions not by executing predetermined code branches but by composing capabilities dynamically, making anticipatory specification of all possible harmful actions practically infeasible. Second, opacity: deep neural networks are notoriously resistant to interpretation. Fraser and Snoswell have demonstrated that this opacity creates severe evidentiary difficulties in tort litigation; causation, the cornerstone of civil liability, becomes difficult to establish when neither plaintiff nor defendant can meaningfully explain why the system acted as it did [4].

Third, distributed agency: production agentic systems increasingly involve orchestrators that delegate subtasks to specialised sub-agents, which may themselves invoke further agents. This nested delegation diffuses causal contribution across multiple technical components and potentially multiple legal entities [5]. Fourth, post-deployment adaptation: unlike conventional software products, which are stable after release, agentic systems may undergo continuous refinement through fine-tuning, retrieval-augmented generation updates, or prompt

injection. The system that causes harm may be meaningfully different from the system originally evaluated and authorised.

III. THE FAILURE OF TRADITIONAL LIABILITY FRAMEWORKS

A. *Fault-Based (Negligence) Liability*

The dominant model in civil liability requires the plaintiff to establish a duty of care, breach of that duty through unreasonable conduct, and causal connection to the loss suffered. Each element presents serious difficulties for agentic AI harm. The duty inquiry has not historically extended in a clear manner to complex software systems operating autonomously, and the relevant standard of reasonable care for novel agentic deployments is undefined. More fundamentally, establishing breach requires showing that the defendant fell below an objective benchmark, but courts have no established metric against which to assess design or deployment choices for systems that have no stable, codified behaviour [6].

Causation is even more problematic. Traditional but-for causation requires the plaintiff to show that harm would not have occurred absent the defendant’s breach. Where an agentic system generates a harmful action through an opaque, emergent process, reconstructing the but-for counterfactual may be practically impossible without access to model weights, training data, and runtime logs—access that defendants routinely deny on grounds of trade secrecy [4].

B. *Product Liability*

Product liability regimes bypass the fault inquiry: the manufacturer of a defective product bears strict liability for harm caused by that defect. The European Product Liability Directive (as revised in 2024) explicitly extends its scope to AI systems and software. However, product liability is premised on a static conception of a product: something manufactured, placed on the market, and thereafter unchanging in its essential character. Agentic systems adapt post-deployment. As De Bruyne, Van Gool, and Gils have argued, it is uncertain whether a system modified through continued training or contextual adaptation remains the same “product” for liability purposes [7].

C. *Guardian (Custodian) Liability*

In civilian legal traditions (French, Québec, Belgian civil law) the guardian of a “thing” bears strict liability for harm caused by that thing. This regime has been proposed as a natural candidate for AI liability [8]. However, the guardian concept presupposes singular, stable custody of a physical object. Agentic AI systems are typically licensed rather than owned, deployed via cloud APIs, and may be simultaneously operated by thousands of deployers. Determining who exercises actual custody—developer, infrastructure provider, or deployer, is a legal question for which existing doctrine offers insufficient guidance.

D. *The Responsibility Gap*

The cumulative effect of these inadequacies is what Asaro terms the “responsibility gap”: the structural possibility that an AI system causes serious harm yet no legal actor bears responsibility commensurate with the gravity of that harm [1]. Gabison and Xian confirm that this gap is not merely theoretical but also technical [9]. The EU Parliament’s 2020 Report on civil liability for AI likewise acknowledged that neither strict nor

fault-based liability, standing alone, is adequate for autonomous AI systems [12].

IV. THE DEPLOYER AS PRIMARY LOCUS OF CIVIL LIABILITY

We argue that the entity best positioned to bear primary civil liability for agentic AI harm is the deployer, the legal person who puts an agentic system into operational use in a specific context and for a specific purpose. This claim rests on four grounds.

First, contextual control. While developers design the system’s underlying capabilities, deployers determine the deployment context: the tasks the system is authorised to perform, the data it may access, the users it interacts with, and the safeguards surrounding its operation. The harm-causing potential of an agentic system is in large measure a function of deployment choices. Barrett et al.’s STAMP/STPA analysis of AI loss-of-control incidents identifies inadequate deployment constraints as a dominant causal factor in unsafe AI behaviour, more frequently implicated than underlying model deficiencies [10].

Second, ongoing monitoring capacity. Unlike developers, who lose operational visibility after system release, deployers have continuous access to system behaviour logs, user interaction data, and performance metrics. This information asymmetry supports imposing liability on the party with the greater ability to detect and respond to emerging risks. Third, economic benefit: the deployer captures the primary economic surplus from the system’s operation. The principle that liability should follow profit *respondeat superior* in its civilian expression, supports attaching civil responsibility to the deployer [8].

Fourth, the principal-agent analogy. Chopra and White have argued that the relationship between a deployer and an agentic system closely resembles a principal-agent relationship in contract law: the deployer sets goals and delegates their execution to the agent [11]. Under this framework, the deployer bears the legal consequences of the agent’s actions just as a principal bears responsibility for acts performed within the scope of an agent’s authority. This framework is consistent with the EU AI Act (2024), which distinguishes between providers (developers) and deployers, imposing on the latter distinct obligations including conformity assessments, fundamental rights impact assessments, and incident reporting for high-risk AI systems.

V. LIABILITY AS A REQUIREMENT: A SPECIFICATION FRAMEWORK

The preceding analysis yields a normative insight with direct implications for requirements engineering: civil liability obligations are, in effect, legal requirements that must be specified and validated across the AI deployment lifecycle. Reformulating them as requirements rather than ex post legal sanctions creates a constructive link between legal accountability and trustworthy system design. We propose four categories of deployer-oriented legal requirements, each with a legal expression and a corresponding technical design implication.

A. R1 - Duty to Monitor: Traceability Requirement

Legal expression: Deployers must maintain complete and temporally ordered logs of all consequential agentic system actions, including tool calls, API interactions, data accesses, and generated outputs. Logs must be retained for a period sufficient to support civil proceedings and must be producible as evidence upon request. Technical implication: Systems must be designed with instrumented execution environments; logging must be architecturally enforced rather than optionally enabled, and cryptographic integrity mechanisms should prevent post-hoc tampering. This requirement directly addresses the causation evidentiary problem identified in Section III.

B. R2 - Duty of Transparency: Explainability Requirement

Legal expression: Where an agentic system makes or contributes to a decision that causes harm, the deployer must be able to provide an explanation sufficient to establish or rebut the causal link between the system’s action and the harm. This does not mandate full model interpretability; it requires that the deployer maintain the technical means to reconstruct the decision context at any given moment. Technical implication: Systems should implement decision-point logging capturing the relevant context at each consequential action, including retrieved documents in RAG pipelines and tool-selection rationale in tool-use pipelines [4].

C. R3 - Duty of Risk Assessment: Pre-Deployment Specification

Legal expression: Before deploying an agentic system in any context involving potential physical, psychological, financial, or reputational harm to third parties, the deployer must conduct and document a risk assessment that identifies foreseeable harm scenarios, assesses their probability and magnitude, and specifies the mitigations implemented. Technical implication: Risk assessments should be structured using safety engineering methodologies such as STAMP/STPA [10], adapted to the characteristics of AI systems. The output of the risk assessment should generate a set of system-level safety requirements verified before deployment.

D. R4 - Duty to Ensure Compensation: Coverage Requirement

Legal expression: Deployers of agentic systems whose risk profile exceeds a defined threshold must maintain liability insurance or contribute to a sector-level compensation fund sufficient to cover foreseeable harms. This addresses the practical concern that many deployers, including SMEs, may be judgment-proof. Technical implication: Actuarial pricing of AI liability insurance depends on standardised risk metrics, requiring agreement on measurable proxies for harm probability and severity—a research priority for the requirements engineering community.

VI. DISCUSSION

The framework has several important limitations. First, the deployer category is not always clear-cut. In complex agentic architectures, an enterprise may simultaneously be the developer of a fine-tuned model, the deployer of that model, and the operator of a multi-agent orchestration layer. The EU AI Act’s treatment of “providers that also deploy” offers a partial

template for boundary rules, but further specification is needed. Second, foundation model developers are not absolved: where harm is attributable to a deficiency in the underlying model that the deployer could not reasonably have detected or mitigated, a residual product liability claim against the developer should be preserved. The deployer hypothesis proposes a rebuttable presumption, not an exclusive liability channel.

Third, cross-jurisdictional harmonisation remains a challenge. The framework draws primarily on European civil law traditions. Common law jurisdictions, including Canada and Australia, have developed distinct negligence doctrines with different burden-allocation rules. The British Columbia Law Institute’s 2023 Consultation Paper on AI and Civil Liability documents the specific difficulties that Canadian tort law faces in accommodating autonomous AI harm, around foreseeability and the standard of care [15]. Aligning these traditions will require sustained comparative law work. Fourth, the framework does not address criminal liability, which raises distinct questions about *mens rea* and organisational culpability.

Despite these limitations, redefining civil liability as a requirements engineering challenge offers a productive research agenda for the scientific community. The focus on identifying and specifying requirements for trustworthy AI aligns perfectly with the regulatory work needed to translate operators’ obligations into verifiable and testable system requirements. Lin and Wu’s analysis of AI liability reform in China similarly argues that mandatory requirements for evidence disclosure—a specific form of our traceability requirement R1—are necessary to address the evidentiary asymmetry inherent in AI systems [13]. De Bruyne and Ooms reach similar conclusions in the European context [14].

VII. CONCLUSION

The agentic AI revolution has outpaced the legal frameworks designed to govern its consequences. Traditional tort and product liability regimes fail systematically when applied to autonomous systems whose harmful behaviour is emergent and causally distributed across complex supply chains. This paper has argued that the deployer—as the entity that controls deployment context, benefits from the system’s operation, and retains ongoing monitoring capacity—should bear primary civil liability for agentic AI harm.

More broadly, we have proposed that civil liability obligations should be reconceived as legal requirements: specifiable, verifiable constraints on system design and operation that drive trustworthiness into the AI deployment lifecycle. The four deployer requirements: traceability (R1), explainability (R2), pre-deployment risk assessment (R3), and compensation coverage (R4), translate legal duties into concrete engineering specifications. The requirements engineering community is uniquely positioned to close the gap between legal aspiration and technical implementation. Developing standard methods for liability-compliant AI design is not merely an academic exercise; it is a prerequisite for the trustworthy integration of agentic AI into social and institutional life.

VIII. REFERENCES

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